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Does kinematic gait quality improve with functional gait recovery? A longitudinal pilot study on early post-stroke individuals

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ABSTRACT

Typical clinical gait outcomes mostly focus on function; only sparse information exists on gait quality, i.e. symmetry or more natural gait patterns. It remains unclear whether functional gait recovery improves with gait quality, or whether these are two independent processes. The objective of this observational pilot study is to examine whether the gait quality improves with gait function (i.e. speed) over the course of early recovery. Full lower body gait kinematics were measured longitudinally in a clinical environment using wearable inertial measurement units. We recorded six individuals with subacute stroke (<1 month) for a total of 56 physical therapy sessions over the initial recovery stage of 12 weeks. We examined relations between gait symmetry in spatiotemporal, limb and joint kinematic parameters compared to gait function. We observed that overall gait symmetry improved with walking speed, but limb and joint kinematic parameters remained asymmetric at the maximum level of recovery (both $p < 0.01$). We also found that limb kinematic parameters ($R^2 = 41.9\%$) of the impaired side was preferentially associated with functional gait recovery over joint kinematics ($R^2 = 33.1\%$). These data suggest that our pilot cohort did not achieve “true” gait recovery despite achieving typical measures of recovery in gait speed and spatiotemporal symmetry. These initial results illustrate the multifaceted nature of recovery and justify further research on monitoring gait quality with a larger clinical study, providing insight for more effective training regimens.

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1. Introduction

Restoration of gait and motor recovery following neurological injury such as stroke involves improvements both in functional ability and quality of movements. However, typical clinical outcomes for measuring gait are mostly focused on functional ability such as gait speed (Tilson et al., 2010), timed-up-and-go (Podsiadlo and Richardson, 1991), six minute walk (Enright, 2003), and others, whereas gait quality is ignored. Disturbances in gait quality are associated with an increased risk of falls (Weerdesteyn et al., 2008), greater energy expenditure (Finley et al., 2013) and may lead to long-term problems such as learned non-use or use-dependent plasticity and pain (Jones, 2017; Levin et al., 2009). For instance, patients with asymmetric gait have shown greater metabolic energy expenditure from 50% to 67% more than that of body weight-matched healthy controls with same walking speed (Woolley, 2001).

Parameters to measure gait quality can be broken down into more than just spatiotemporal characteristics (i.e., step length or step time). Spatiotemporal symmetry has been well charted in post-stroke individuals (Balasubramanian et al., 2007; Patterson et al., 2015, 2010). However, stroke survivors exhibit significant asymmetry in joint kinematics with greater inter-individual variability than spatiotemporal characteristics (Woolley, 2001). Typical asymmetry in joint kinematics includes reduced hip extension, knee flexion and ankle dorsi/plantarflexion, and knee hyperextension on the impaired side (Kim and Eng, 2004; Perry and Burnfield, 1992). While these studies have attempted to characterize disturbances in joint kinematics after stroke, the majority report instantaneous or short-term measurements on chronic patients (Kim and Eng, 2004; Woolley, 2001). There is currently a lack of information regarding how these joint kinematics change over the course of recovery, especially the critical subacute phase (<3 months).

Limb kinematics, used to describe end-effector (i.e. foot) motion in task space, can be used to characterize locomotor function. For instance, Chang et al. suggested that joint kinematics were

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coordinated with greater variance to preferentially conserve and stabilize the limb kinematics of walking after neuromuscular injury of cats (Chang et al., 2009). It is reflective of the concept of an uncontrolled manifold hypothesis, where control degrees-of-freedom are reduced by allowing variability in uncontrolled variables (Scholz and Schöner, 1999). In humans, however, no evidence exists to show how spatiotemporal characteristics, along with limb or joint kinematics, change over the course of early recovery and whether they correspond with gait function. Characterizing how these gait parameters change over recovery can help determine how to time interventions to maximize function.

The objective of this observational study is to examine changes in gait quality in terms of spatiotemporal, limb and joint kinematics over the initial stage of recovery. The primary technical challenge of this endeavor is measuring kinematics in a clinical environment longitudinally. We overcame this problem by using portable, unobtrusive inertial motion capture to measure gait kinematics over the initial recovery stage of 12 weeks. Since asymmetry in joint kinematics shows greater variability than spatiotemporal characteristics (Woolley, 2001), we expected that gait quality will remain asymmetric over the course of early recovery, driven by joint kinematics. Secondly, we also expected that in the impaired side, limb kinematic parameters will associate with gait speed more than joint kinematics, suggesting an uncontrolled manifold in the joint kinematics domain.

This preliminary study describes a novel characterization of disturbances in gait quality during the early recovery process after stroke. The information from this study details the importance of gait quality, it helps delineate underlying kinematic mechanisms of gait recovery, and provides insight for more effective therapy regimens.

2. Methods

2.1. Participants

We recruited nine individuals with subacute stroke (<1 month) to participate in this study approved by the University of Texas Institutional Review Board and St. David's Medical Center, Austin, TX. Among nine patients, six individuals were longitudinally monitored for a total of 56 physical therapy sessions consisting of gait-related activity. Individuals ranged in age and impairment level (see Table 1). Inclusion criteria of this study were: ischemic cerebral infarction based on magnetic resonance imaging (MRI) data, hemiparesis, premorbidly independent, within two weeks following injury or as soon as able to participate in gait training determined by the physical therapist. Exclusion criteria include cerebellar damage, prior stroke, stroke-related pain syndromes, or functionally relevant neuromuscular impairments. Participants were allowed to use any assistive devices including a wheelchair, body weight support, walker, cane or ankle foot orthosis (AFO)

during sessions if necessary (see Table 1). Prior to the first session, the experimenter and therapist explained all the experimental procedures to each participant and obtained informed consent. We began monitoring as soon as participants were capable of beginning gait therapy, i.e., as soon as walk independently without therapist's full assistance. Three participants were excluded from the dataset because two of them were not able to walk before discharge and one discharged before the recording session. Three of the participants included in this analysis completed less than the full 12 recording sessions. One participant was discharged to a different hospital (P2), one lacked sufficient insurance coverage (P3), and one was discharged to home therapy (P4).

2.2. Experimental setup and protocol

Our goal was to consecutively monitor gait kinematics during physical therapy sessions. We targeted a maximum of 12 recording sessions over the course of the first three months throughout the inpatient to outpatient phases. Our recording session frequency was greater in the earlier stages of recovery, i.e. two/week for three weeks, one/week for four weeks, then one every other week. For each session, we attached seven commercial inertial measurement unit (IMU) motion sensors (XSens, Enschede, The Netherlands) on the pelvis and bilaterally on thighs, shanks and feet using custom-made elastic bands. The anthropometric measurements were estimated by height of each subject and used as input of the IMU software (Winter, 2009). IMUs measured three-dimensional orientations of all lower body segments as well as joint angle trajectories at a sampling rate of 60 Hz. Data was transmitted wirelessly to a commercial Latitude E5470 laptop (Dell Computers, Round Rock, TX).

The total setup time took approximately 10–20 min. All setup was performed immediately prior to the normally scheduled physical therapy sessions. After donning sensors, participants were asked to maintain a straight standing position for five seconds to calibrate the system as recommended by proprietary IMU manufacturer. Therapist assistance was provided for this calibration as necessary. Following calibration, participants performed typical 1-hour training under the supervision of a physical therapist incorporating a number of tasks including overground and treadmill walking. The experimenter used a cart to follow the participant and therapist to maintain wireless communication of sensor signals (approximately 3–5 m away from the participant). After the session, all sensors were removed from the participant.

2.3. Gait features and outcomes

On average, approximately 45.5% of the duration of the 1-hour therapy session was dedicated to gait training. For each session, we sampled normal, straight overground walking (between 5 and 80 strides depending on participant's severity) at a comfortable speed to extract gait quality measures. Custom software was written in

Table 1
Demographics and clinical characteristics of participants.

Patient #	P1	P2	P3	P4	P5	P6
Age (years)	61	59	68	68	52	27
Sex	M	M	M	M	F	M
BMI (kg/cm ²)	29.6	25.8	28.7	28.2	25.8	40.5
Affected side	L	L	R	R	L	R
1st recording after admission (days)	15	5	6	7	20	31
Recordings (sessions)	12	5	10	5	12	12
Ankle foot orthosis (used sessions)	–	5	10	–	–	12
Assistive device (used sessions)	1	5	3	5	10	11
Baseline impairment (admission motor FIM score)	18	26	31	22	22	16

BMI – body mass index, FIM – functional independence measure, M – male, F – female, L – left, R – right.

MATLAB (Mathworks, Inc. R2016a, Natick, MA) to calculate the outcomes. The joint angle trajectories of the sampled gait portion were truncated and normalized into 100% of gait cycle from heel strike to heel strike of the left foot (Zeni Jr. et al., 2008). Fig. 1 illustrates an example of recorded gait kinematics data from a session including bilateral joint angle trajectories of single gait cycle in sagittal plane. Outcomes were extracted by imposing the gait kinematics data into a lower body kinematic model with the lower limb lengths estimated by the anthropometric data based on height of each individuals as described below (Winter, 2009).

Spatiotemporal parameters included step length (SL) and step time (ST), defined as the linear distance between right and left feet, and the duration of each step, respectively (Fig. 2, left). Limb kinematic parameters incorporated limb length (LL) (Chang et al., 2009), leg extension angle (LEA) (Peterson et al., 2010), and foot path area (FPA) (Banala et al., 2009) defined as the range of linear distance between hip and foot, the angle between a line from hip to the foot and vertical before toe-off, and area of foot pattern from hip in sagittal plane during gait cycle, respectively (Fig. 2, middle). Joint kinematic parameters were defined as the range of motion (RoM) from selected joint variables, including all three rotations of hip, knee flex/extension and ankle dorsi/plantarflexion (Fig. 2, right). All parameters involved affected and unaffected sides, enabling symmetry analysis.

We selected gait speed as the main outcome measure representative of functional gait recovery (Holleran et al., 2014; Hornby et al., 2016). The number of strides for each session was varied (between 5 and 80 strides) due to different impairment levels within and between subjects. In each session, the average gait speed was calculated by dividing average stride distance by the average time of gait cycles. A simple correlation analysis showed a strong association between calculated gait speed and clinical outcome measure, the six-minute walk test ($r = 0.91$).

2.4. Gait asymmetry measures

2.4.1. Overall gait asymmetry

Gait symmetry, and its inverse, asymmetry, are common measures of gait quality (Ramakrishnan et al., 2018; Viteckova et al., 2018). In its simplest form, symmetry is represented by the ratio of stance time on each limb or step length (Balasubramanian et al., 2007). However, spatiotemporal asymmetry does not necessarily equate with asymmetry in other domains such as kinematics. Here we combined all the aforementioned gait parameters into one asymmetry measure by employing the combined gait asymmetry metric (CGAM) (Ramakrishnan et al., 2018). The CGAM combines different asymmetry parameters using the Mahalanobis distance. Mahalanobis distance is a distance measure of a

multi-dimensional data point in a data distribution space (De Maesschalck et al., 2000). Compared to Euclidean distance, the Mahalanobis distance takes covariance of data into account to standardize the scaling of distance measure in multi-dimensional space. The CGAM uses the Mahalanobis distance from ideal symmetry (i.e., zero) to the asymmetry ratio obtained from gait parameters. For each session, the asymmetry ratio of i^{th} gait parameter was calculated with the asymmetry ratio index given by

$$|ar_i| = \frac{|US_i - AS_i|}{0.5(US_i + AS_i)} \quad (1)$$

where US_i and AS_i are the i^{th} gait parameter of unaffected and affected sides, respectively, and i represents each gait parameter. Note that the asymmetry ratio, $|ar_i| = 0$, has no asymmetry. The asymmetry of parameters from spatiotemporal, limb and joint kinematic domains was then evaluated with the Mahalanobis distance, D_m , given by

$$D_m = \sqrt{\frac{AR * S^{-1} * AR^T}{\sum (S^{-1})}} \quad (2)$$

where $AR = [|ar_1|, \dots, |ar_n|]$ is the vector of asymmetry ratios, $S \in \mathbb{R}^{n \times n}$ is the covariance matrix of asymmetry ratio data, and n is the number of gait parameters. $\sum (S^{-1})$ is the summation of the inverse covariance matrix to provide more of a weighted means approach to decrease the variability of results for the same gait parameters (Ramakrishnan et al., 2018). As a basis for comparison, we extracted the symmetric threshold based on the AR of our gait database (Yun et al., 2014) obtained from 113 healthy individuals walking at 3 km/h (Table 2). The symmetric threshold for CGAM was then calculated by combining those symmetric threshold values of input gait parameters, and defined as “symmetric” walking if the Mahalanobis distance, D_m , was less than the symmetric threshold of CGAM.

2.4.2. Asymmetry of individual gait parameters

The asymmetry of individual parameters in spatiotemporal (SL, ST), limb (LL, LEA, FPA) and joint kinematic (RoMs) domains was then independently analyzed to explore the main contributor of overall gait asymmetry. AR was used to evaluate asymmetry of individual parameters. Note that with the absolute value in Eq. (1), the asymmetry ratio only indicates whether each parameter is asymmetric or not, without taking into account the direction of which side is greater. Similar to overall asymmetry, an asymmetry ratio less than the symmetric threshold value obtained from Table 2 was considered “symmetric” walking for the individual parameters.

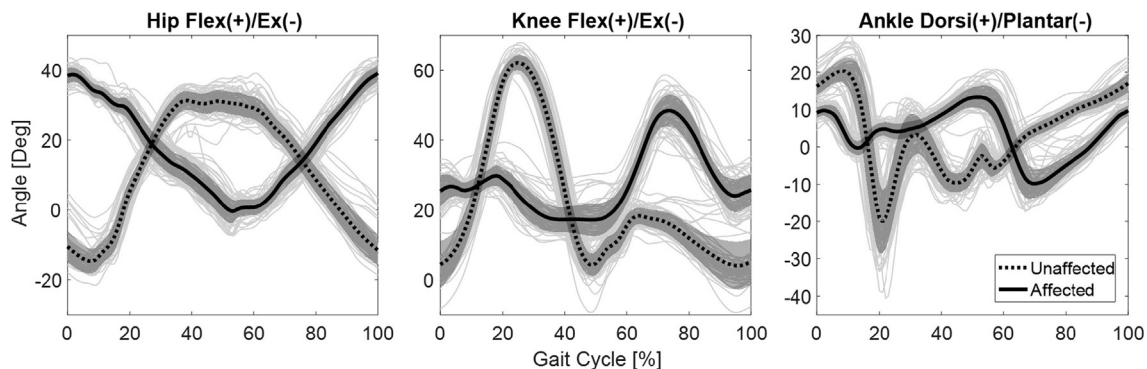


Fig. 1. Example joint angle trajectories of single gait cycle (left heel strike to heel strike) in sagittal plane (12th session of patient 5), hip flexion/extension (left), knee flexion/extension (middle), and ankle dorsi/plantarflexion (right).

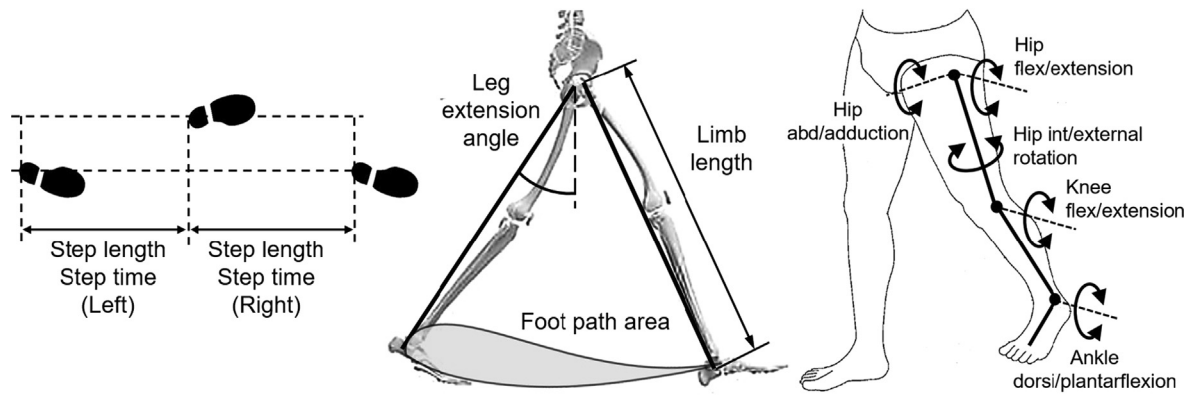


Fig. 2. Gait quality domains and their parameters. Spatiotemporal characteristics include step length and step time (left), limb kinematics include leg extension angle, limb length and foot path area (middle), and joint kinematics include range of selected joint motions (right).

Table 2

Symmetric threshold of gait parameters from gait database of 113 healthy individuals walking at 3 km/h.

Domain	Gait parameters	$ r_h (\%)$
Spatiotemporal	Step length	5.4
	Step time	6.1
Limb kinematics	Leg extension angle	0.6
	Limb length	2.5
	Foot path area	2.9
	Hip Flex/Ex	1.9
Joint kinematics	Hip Abd/Add	0.1
	Hip Int/Ext	1.2
	Knee Flex/Ex	2.7
	Ankle Dorsi/Plantar	7.8

Flex/Ex – flex/extension, Abd/Add – abd/adduction, Int/Ext – int/external rotation, Dorsi/Plantar – dorsi/plantarflexion, $|r_h|$ – symmetry ratio threshold of individual gait parameters from healthy subjects.

2.4.3. Limb vs. joint kinematics

Each limb and joint kinematic parameter was transformed into a common space (i.e., domain) using the Mahalanobis distance metric, which was used in the overall asymmetry test (i.e., CGAM). In the limb vs. joint kinematics test, limb kinematic parameters (LL, LEA, FPA) and joint kinematic parameters (RoMs) were normalized and then combined with the Mahalanobis distance, respectively, in order to produce parameters from each limb and joint kinematic domain into one value per session. For each side, we examined how changes in each kinematic domain associated with changes in functional gait recovery (i.e., gait speed).

2.5. Statistical analysis

R version 3.4.1 (2017 The R foundation for Statistical Computing) was used for the statistical analysis. For the analysis, 53 sessions from a total of 56 recording sessions were used, excluding outliers (8th session of P3, 9th session of P5, 2nd session of P6) with an exaggerated range of motion ($<60^\circ$) at ankle dorsi/plantarflexion (e.g., distorted by magnetic interference of the IMUs).

A linear mixed regression model was used to evaluate the relationship between asymmetry of gait parameters and speed as a functional gait measure with significance level of $\alpha < 0.05$. For the overall gait asymmetry test, the Mahalanobis distances of spatiotemporal, limb and joint kinematics were standardized and used as dependent variables. For the asymmetry test of individual parameters, the asymmetry ratio, $|ar_i|$, of each gait parameter was used as a dependent variable. For both tests, gait speed and subject were used as fixed and random effects of the mixed model, respectively. In addition, a one-sided one-sample *t*-test was used to

evaluate the asymmetry of overall and individual gait parameters at the maximum functional recovery with significance level of $\alpha < 0.05$. The asymmetry ratio at maximum gait speed of each patient was extracted and compared with the symmetric threshold value for each gait parameter from Table 2.

For the limb vs. joint kinematics test, the linear mixed regression model was used to evaluate the relations between parameters of both kinematic domains and gait speed with significance level of $\alpha < 0.05$. Coefficient of determination (R^2) (Nakagawa and Schielzeth, 2013), Akaike information criteria (AIC), and Bayesian information criteria (BIC) were used to compare goodness-of-fit of the models.

3. Results

3.1. Asymmetry of overall gait parameters

We first examined the relations of overall asymmetry over spatiotemporal, limb and joint kinematic domains and gait speed over the initial recovery phase of three months based on the CGAM. Fig. 3 illustrates changes in the Mahalanobis distances of spatiotemporal (left), limb (middle), and joint (right) kinematics across changing gait speeds, respectively. The results indicated that the asymmetry improved significantly in spatiotemporal ($\beta = -0.47$, $p < 0.01$) and limb kinematic parameters ($\beta = -0.36$, $p < 0.05$) whereas joint kinematics ($\beta = -0.25$, $p = 0.14$) revealed an improving trend. At the session with maximum functional recovery (maximum speed), spatiotemporal parameters ($D_{max} = 0.11$, $p = 0.11$) did not reveal significant asymmetry but limb ($D_{max} = 0.56$, $p < 0.001$) and joint ($D_{max} = 0.38$, $p < 0.01$) kinematic parameters remained significantly different (asymmetric) compared to the symmetric threshold. All results are summarized in Table 3.

3.2. Asymmetry of individual gait parameters

According to the overall asymmetry test, all gait parameters revealed improving trends but both limb and joint kinematic parameters remained asymmetric at the maximum functional recovery. Thus, we used linear mixed regression models and one-sided one-sample *t*-tests on individual gait parameters independently to further investigate major contributors of overall gait asymmetry. The results revealed that the asymmetry improved significantly at SL ($\beta = -0.33$, $p < 0.05$), LEA ($\beta = -0.65$, $p < 0.01$), and hip flex/extension ($\beta = -0.15$, $p < 0.05$). At maximum gait speed, all limb kinematic parameters including LEA ($|ar_{max}| = 0.29$, $p < 0.05$), LL ($|ar_{max}| = 0.62$, $p < 0.01$), and FPA ($|ar_{max}| = 0.87$, $p < 0.01$) and joint kinematic parameters including

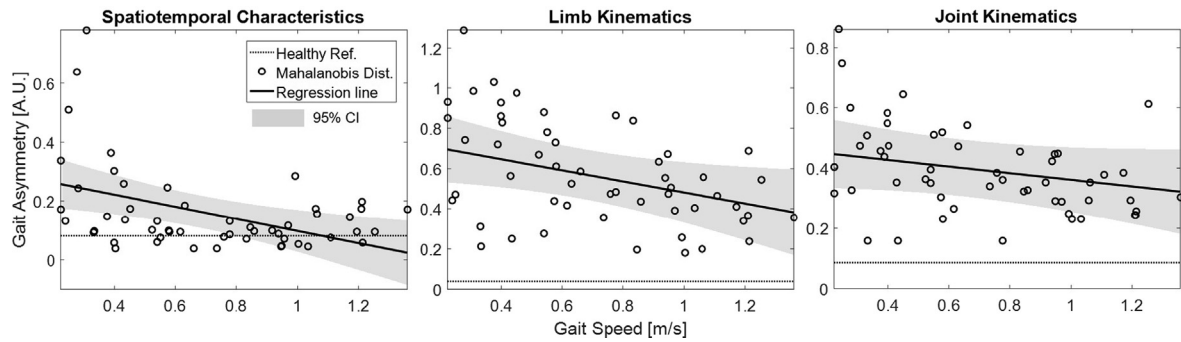


Fig. 3. Change in overall gait symmetry measures based on combined gait asymmetry metric (CGAM) including spatiotemporal characteristics (left), limb kinematics (middle), and joint kinematics (right) with gait speed. The dotted line is considered as the threshold for symmetric walking. The regression lines are based on linear mixed regression model (Table 2).

Table 3

Results of a linear mixed regression model based on association between combined gait asymmetry metric (CGAM, dependent variable) and gait speed (independent variable). The table also includes results of a one-sided, one-sample *t*-test of CGAM determining gait asymmetry at maximal gait speed (i.e. functional recovery).

Domain	Linear mixed regression		One-sided one sample <i>t</i> -test	
	β [95% CI]	<i>p</i> -value	D_{max} [95% CI]	<i>p</i> -value
Spatiotemporal	−0.47 [−0.78, −0.15]	< 0.01	0.11 [0.07, inf]	0.11
Limb kinematics	−0.36 [−0.68, −0.04]	< 0.05	0.56 [0.41, inf]	< 0.001
Joint kinematics	−0.25 [−0.59, 0.09]	0.14	0.38 [0.27, inf]	< 0.01

CI – confidence interval, β – slope of regression, D_{max} – Mahalanobis distance at maximum functional gait recovery (max speed)

hip flex/extension ($|ar_{max}| = 0.14$, $p < 0.05$), hip abd/adduction ($|ar_{max}| = 0.29$, $p < 0.05$), hip int/external rotation ($|ar_{max}| = 0.36$, $p < 0.01$), and knee flex/extension ($|ar_{max}| = 0.27$, $p < 0.05$) remained significantly different (i.e. asymmetric) compared to the symmetric threshold (Table 4). As expected, both spatiotemporal parameters (SL and ST) did not show significant differences from the symmetric threshold.

3.3. Limb kinematics vs. joint kinematics

We examined whether limb kinematic parameters were preferentially associated with functional gait recovery compared to joint kinematics, consistent with the uncontrolled manifold hypothesis (Chang et al., 2009). For each side, we transformed both limb and joint kinematic parameters into a common space by combining all parameters of each kinematic domain using the Mahalanobis distance. A simple correlation analysis between limb and joint kinematic domains revealed $r = 0.67$ and $r = 0.38$ for unaffected and affected sides, respectively. The results revealed that all kinematic parameters in all domains of both sides significantly

associated with gait speed (all $p < 0.001$). However, according to the goodness-of-fit measures, the variance accounted for of limb kinematic parameters ($R^2 = 41.9\%$) was greater than in joint kinematics ($R^2 = 33.1\%$) at the affected side, whereas unaffected side showed a greater correlation in joint kinematics ($R^2 = 57.6\%$) than limb kinematics ($R^2 = 27.5\%$) with consistent trends in AIC and BIC (Table 5).

4. Discussion

Typical clinical outcomes focus more on gait function than quality (Hornby et al., 2016), resulting in very little knowledge of how gait quality develops over the course of recovery. The primary goal of this study was to establish initial evidence regarding measurements of changes in gait quality over the early stage of recovery. We observed that asymmetry of gait parameters overall improved with gait function, but both limb and joint kinematics remained asymmetric at the maximum functional recovery. Second, leg extension and hip flex/extension were most associated with

Table 4

Results of linear mixed regression models with asymmetry ratio of individual gait parameters (dependent variables) and gait speed (independent variable). Additional one-sided, one-sample *t*-tests of gait asymmetry ratio at maximal gait speed are also included.

Domain	Gait parameters	Linear mixed regression		One-sided one sample <i>t</i> -test	
		β [95% CI]	<i>p</i> -value	$ ar_{max} $ [95% CI]	<i>p</i> -value
Spatiotemporal characteristics	Step length	−0.33 [−0.60, −0.06]	< 0.05	0.16 [0.04, inf]	0.07
	Step time	−0.06 [−0.16, 0.04]	0.26	0.07 [0.01, inf]	0.43
Limb kinematics	Leg extension angle	−0.65 [−1.05, −0.24]	< 0.01	0.29 [0.03, inf]	< 0.05
	Limb length	−0.09 [−0.40, 0.22]	0.55	0.62 [0.29, inf]	< 0.01
	Foot path area	−0.50 [−1.04, 0.04]	0.07	0.87 [0.38, inf]	< 0.01
Joint kinematics	Hip Flex/Ex	−0.15 [−0.30, −0.00]	< 0.05	0.14 [0.06, inf]	< 0.05
	Hip Abd/Add	−0.06 [−0.36, 0.23]	0.68	0.29 [0.03, inf]	< 0.05
	Hip Int/Ext	−0.02 [−0.27, 0.24]	0.90	0.36 [0.20, inf]	< 0.01
	Knee Flex/Ex	−0.22 [−0.45, 0.00]	0.05	0.27 [0.03, inf]	< 0.05
	Ankle Dorsi/Plantar	−0.17 [−0.43, 0.08]	0.18	0.30 [0.08, inf]	0.05

Flex/Ex – flex/extension, Abd/Add – abd/adduction, Int/Ext – int/external rotation, Dorsi/Plantar – dorsi/plantarflexion, β – slope of regression, $|ar_{max}|$ – asymmetry ratio at maximum functional gait recovery (max speed).

Table 5
Results of linear mixed regression models with Mahalanobis magnitude of limb and joint kinematic parameters at unaffected and affected sides (dependent variables) and gait speed (independent variable).

Side	Kinematic domain	Linear mixed regression			
		Slope [95% CI]	R^2	AIC	BIC
Unaffected side	Limb kinematics	0.53 [0.26, 0.81] ^{***}	27.5%	139.7	147.6
	Joint kinematics	0.77 [0.56, 0.97] ^{***}	57.6%	111.0	118.8
Affected side	Limb kinematics	0.79 [0.49, 1.10] ^{***}	41.9%	131.4	139.2
	Joint kinematics	0.58 [0.35, 0.81] ^{***}	33.1%	135.7	143.6

^{***} $p < 0.001$, R^2 – coefficient of determination, AIC – Akaike information criteria, BIC – Bayesian information criteria.

improved asymmetry, but many gait parameters remained asymmetric at the maximum functional recovery. Third, limb kinematics of the impaired side was preferentially associated with functional gait recovery over joint kinematics. These results offer preliminary evidence justifying the importance of gait quality in stroke recovery and indicate a specific set of parameters that may be more critical than others in the early recovery process.

Asymmetry in kinematics is often underreported in literature, perhaps due to the difficulties in measuring kinematic data in clinical settings. Based on our longitudinally measured gait kinematics data from wearable IMUs, the spatiotemporal as well as kinematic asymmetries were captured and evaluated along with the improved gait speed over the course of early recovery. The results of our overall asymmetry test indicated that all measures of asymmetry improved with gait speed (Fig. 3). This result agrees with earlier work that found faster and more symmetric walking showed lower metabolic cost than merely walking faster or more symmetrically (Awad et al., 2015).

We observed that leg extension angle and hip flex/extension asymmetry measures were most associated with gait speed. This suggests that the leg extension and hip flex/extension may be important predictors of functional gait recovery. These results agree with previous studies that found leg and hip extension were significantly associated with propulsive forces during walking as well as gait speed (Kim and Eng, 2004; Peterson et al., 2010). However, all parameters other than ankle dorsi/plantarflexion remained significantly asymmetric compared to the healthy symmetric threshold at the maximum gait speed (Table 4). This could mean that the underlying control system involved in gait recovery prioritizes motion at the hip and leg extension over other gait parameters. It may also suggest that there is still a room to improve on those asymmetric gait parameters and additional therapeutic efforts need to be made to achieve a complete recovery with symmetric walking.

While all kinematic domains of both sides correlated with gait speed, the results of goodness-of-fit measure revealed that the variance explained by limb kinematic parameters ($R^2 = 41.9\%$) was greater than joint kinematics ($R^2 = 33.1\%$) in the affected side, indicating stronger association in limb kinematics with gait speed over the joint kinematics on impaired side (see Table 5). The goodness-of-fit measure for limb kinematics is similar to a clinically relevant relationship between steps per session and recovery ($R^2 = 42\%$) found by Hornby et al. (2016). We also observed an 8.8% difference between goodness-of-fit measures of limb and joint kinematics. While there is no basis of comparison to judge the magnitude of this difference, the result itself suggests that post-stroke individuals may preferentially coordinate limb function using the limb kinematics and compensate in the joint kinematics. These findings are consistent with the concept of an uncontrolled manifold in joint kinematics space found previously in animal models (Bauman and Chang, 2013; Chang et al., 2009).

In the unaffected side, joint kinematic parameters ($R^2 = 57.6\%$) revealed greater explained variance than limb kinematics

($R^2 = 27.5\%$). These results are contrary to the trends observed in the affected limb and further, would seem contrary to the uncontrolled manifold hypothesis. While feasible that the uncontrolled manifold hypothesis is not an ideal model, this finding may be explained by greater residual function in the limb kinematics in the unaffected limb compared to joint kinematics. Indeed, our additional analysis revealed that the intercept of the unaffected limb kinematics (0.39) was over two times greater than that of the affected side (0.18) (Supplementary Fig. 1). This indicates that a ceiling effect in the limb kinematics of the unaffected side may reduce its explanatory ability. These results may provide insight into how to interpret kinematic biomarkers of recovery.

This study represents a proof-of-concept to show how measures of gait quality associate with changes in gait function in the most critical stage of recovery. However, results need to be interpreted with an important proviso. This study was limited to a small sample size of six patients with a restricted range of impairments (only individuals who were able to walk within one month post-injury). The study was also conducted at the same hospital with similar training techniques. Thus, we cannot generalize these results to a larger population of stroke patients. However, our data show the feasibility of unobtrusively recording gait during therapy longitudinally and offer a framework for how it can help improve our understanding of recovery. We expect this study will justify more expanded research on gait quality with a larger post-stroke population for more reliable generalization.

Another important advance of this work is providing a novel methodology to track gait quality by measuring joint kinematics over the course of the initial recovery stage. However, measuring joint kinematics is relatively more sensitive to errors than other conventional gait measures such as gait speed or spatiotemporal characteristics. Typical drawbacks of IMUs, magnetic distortion (De Vries et al., 2009) and improper calibration in a straight standing posture, may have influenced our results. To mitigate these issues, we used an IMU motion capture system specifically developed for measuring kinematics (Paulich et al., 2018) and validated in a laboratory environment (Lee et al., 2019). Recalibration was performed regularly by the experimenter based on visual inspection, although without any template to compare, it is possible there were some errors in calibration. Given range-of-motion-based outcome measures, it is unlikely that calibration issues substantially affected our results. We expect further advances in motion capture technology will help to measure more accurate kinematics data, providing more detailed and reliable results of characterizing post-stroke gait.

5. Conclusions

The purpose of this observational study was to monitor changes in gait quality with functional gait recovery longitudinally over the initial recovery phase by unobtrusively measuring full lower body kinematics. We observed that while post-stroke individuals

exhibited overall improvements in gait quality with gait function, symmetry did not fully recover. We conclude that using inertial motion capture to longitudinally observe gait quality is feasible and may provide a more holistic representation of recovery. These initial results justify further research on monitoring gait quality with a larger clinical study.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jbiomech.2020.109761>.

References

- Awad, L.N., Palmer, J.A., Pohlig, R.T., Binder-Macleod, S.A., Reisman, D.S., 2015. Walking speed and step length asymmetry modify the energy cost of walking after stroke. *Neurorehabil. Neural Repair* 29, 416–423.
- Balasubramanian, C.K., Bowden, M.G., Neptune, R.R., Kautz, S.A., 2007. Relationship between step length asymmetry and walking performance in subjects with chronic hemiparesis. *Arch. Phys. Med. Rehabil.* 88, 43–49.
- Banala, S.K., Kim, S.H., Agrawal, S.K., Scholz, J.P., 2009. Robot assisted gait training with active leg exoskeleton (ALEX). *IEEE Trans. Neural Syst. Rehabil. Eng.* 17, 2–8.
- Bauman, J.M., Chang, Y.-H., 2013. Rules to limp by: joint compensation conserves limb function after peripheral nerve injury. *Biol. Lett.* 9, 20130484.
- Chang, Y.-H., Auyang, A.G., Scholz, J.P., Nichols, T.R., 2009. Whole limb kinematics are preferentially conserved over individual joint kinematics after peripheral nerve injury. *J. Exp. Biol.* 212, 3511–3521.
- De Maesschalck, R., Jouan-Rimbaud, D., Massart, D.L., 2000. The mahalanobis distance. *Chemom. Intell. Lab. Syst.* 50, 1–18.
- De Vries, W.H.K., Veeger, H.E.J., Baten, C.T.M., Van Der Helm, F.C.T., 2009. Magnetic distortion in motion labs, implications for validating inertial magnetic sensors. *Gait Posture* 29, 535–541.
- Enright, P.L., 2003. The six-minute walk test. *Respir. Care* 48, 783–785.
- Finley, J.M., Bastian, A.J., Gottschall, J.S., 2013. Learning to be economical: the energy cost of walking tracks motor adaptation. *J. Physiol.* 591, 1081–1095.
- Holleran, C.L., Straube, D.D., Kinnaird, C.R., Leddy, A.L., Hornby, T.G., 2014. Feasibility and potential efficacy of high-intensity stepping training in variable contexts in subacute and chronic stroke. *Neurorehabil. Neural Repair* 28, 643–651.
- Hornby, T.G., Holleran, C.L., Hennessy, P.W., Leddy, A.L., Connolly, M., Camardo, J., Woodward, J., Mahtani, G., Lovell, L., Roth, E.J., 2016. Variable intensive early walking poststroke (VIEWS): a randomized controlled trial. *Neurorehabil. Neural Repair* 30, 440–450. <https://doi.org/10.1177/1545968315604396>.
- Jones, T.A., 2017. Motor compensation and its effects on neural reorganization after stroke. *Nat. Rev. Neurosci.* 18, 267.
- Kim, C.M., Eng, J.J., 2004. Magnitude and pattern of 3D kinematic and kinetic gait profiles in persons with stroke: relationship to walking speed. *Gait Posture* 20, 140–146.
- Lee, J., Shin, S.Y., Ghorpade, G., Akbas, T., Sulzer, J., 2019. Sensitivity comparison of inertial to optical motion capture during gait: implications for tracking recovery, in: 2019 IEEE 16th International Conference on Rehabilitation Robotics (ICORR). IEEE, pp. 139–144.
- Levin, M.F., Kleim, J.A., Wolf, S.L., 2009. What do motor “recovery” and “compensation” mean in patients following stroke?. *Neurorehabil. Neural Repair* 23, 313–319.
- Nakagawa, S., Schielzeth, H., 2013. A general and simple method for obtaining R² from generalized linear mixed-effects models. *Methods Ecol. Evol.* 4, 133–142.
- Patterson, K.K., Gage, W.H., Brooks, D., Black, S.E., McIlroy, W.E., 2010. Evaluation of gait symmetry after stroke: a comparison of current methods and recommendations for standardization. *Gait Posture* 31, 241–246.
- Patterson, K.K., Mansfield, A., Biasin, L., Brunton, K., Inness, E.L., McIlroy, W.E., 2015. Longitudinal changes in poststroke spatiotemporal gait asymmetry over inpatient rehabilitation. *Neurorehabil. Neural Repair* 29, 153–162.
- Paulich, M., Schepers, M., Rudigkeit, N., Bellusci, G., 2018. Xsens MTw Awinda: Miniature Wireless Inertial-magnetic Motion Tracker for Highly Accurate 3D Kinematic Applications. Xsens, Enschede Neth.
- Perry, J., Burnfield, J.M., 1992. Gait Analysis: Normal and Pathological Function.
- Peterson, C.L., Cheng, J., Kautz, S.A., Neptune, R.R., 2010. Leg extension is an important predictor of paretic leg propulsion in hemiparetic walking. *Gait Posture* 32, 451–456.
- Podsiadlo, D., Richardson, S., 1991. The timed “Up & Go”: a test of basic functional mobility for frail elderly persons. *J. Am. Geriatr. Soc.* 39, 142–148.
- Ramakrishnan, T., Lahiff, C.-A., Reed, K.B., 2018. Comparing gait with multiple physical asymmetries using consolidated metrics. *Front. Neurobotics* 12, 2.
- Scholz, J.P., Schöner, G., 1999. The uncontrolled manifold concept: identifying control variables for a functional task. *Exp. Brain Res.* 126, 289–306.
- Tilson, J.K., Sullivan, K.J., Cen, S.Y., Rose, D.K., Koradia, C.H., Azen, S.P., Duncan, P.W., Team, L.E.A.P.S. (LEAPS) I., 2010. Meaningful gait speed improvement during the first 60 days poststroke: minimal clinically important difference. *Phys. Ther.* 90, 196–208.
- Viteckova, S., Kutilek, P., Svoboda, Z., Krupicka, R., Kauler, J., Szabo, Z., 2018. Gait symmetry measures: a review of current and prospective methods. *Biomed. Signal Process. Control* 42, 89–100.
- Weerdesteijn, V., de Niet, M., van Duijnhoven, H.J., Geurts, A.C., 2008. Falls in individuals with stroke. *J. Rehabil. Res. Dev.* 45, 1195–1214.
- Winter, D.A., 2009. Biomechanics and Motor Control of Human Movement. John Wiley & Sons.
- Woolley, S.M., 2001. Characteristics of gait in hemiplegia. *Top. Stroke Rehabil.* 7, 1–18.
- Yun, Y., Kim, H.-C., Shin, S.Y., Lee, J., Deshpande, A.D., Kim, C., 2014. Statistical method for prediction of gait kinematics with Gaussian process regression. *J. Biomech.* 47, 186–192.
- Zeni Jr, J.A., Richards, J.G., Higginson, J.S., 2008. Two simple methods for determining gait events during treadmill and overground walking using kinematic data. *Gait Posture* 27, 710–714.